

Conducting polymers. Conducting polymer gas sensors operate based on changes in the electrical resistance caused by adsorption of gases onto the sensor surface.

Piezoelectric sensors. Adsorption of gas molecules onto the surface of the polymer leads to changes in the mass on the sensor surface. This, in turn, produces a change in the resonant frequency of the crystal. This change in frequency is proportional to the concentration of the test material.

Surface acoustic wave (SAW). These are a class of micro-electromechanical systems (MEMSes) that rely on modulation of surface acoustic waves to sense a physical phenomenon.

Optical sensors. These are more complex than typical sensor-array systems having transduction mechanisms based on changes in electrical resistance. Optical sensors work by means of light-modulation measurements.

Cyranose 320

Cyranose 320 is a handheld e-nose developed by Cyranose Sciences of Pasadena, California, USA, in 2000. It is based on sensor research performed by Prof. Nathan Lewis of California Institute of Technology, USA. Applications researched using Cyranose 320 include detection of chronic obstructive pulmonary diseases and other medical conditions as well as industrial applications generally related to quality control or contamination detection.

Cyranose 320 utilises the NoseChip array of nanocomposite sensors and advanced pattern-recognition algorithms to detect and recognise the chemical vapour of interest. In combination, these technologies enable rapid detection and identification of substances based on their respective chemical profiles.

For each method, it provides simple, clear, actionable results

such as Accept or Reject, Mixture Identified or Contaminated. Each measurement is assessed based on a statistical quality rating of one to five stars.

More about the e-nose

It is a cheap and non-destructive instrument that can be operated by non-specialists. As the process is automatic, the cost of each measurement is low.

Measurement cycle of the e-nose should be faster so as to increase throughput. Sensitivity and selectivity of the e-nose is still far from a mammalian nose.

COMPARISON OF E-NOSE WITH BIOLOGICAL NOSE

Biological nose	e-nose
Inhaling	Pump
Mucus	Filter
Olfactory epithelium	Sensors
Binding with proteins	Interaction
Enzymatic proteins	Reaction
Cell membrane depolarised	Signal
Nerve impulses	Circuitry and neural network

The e-nose has applications in the following areas:

Medical diagnosis and health monitoring. This can be further categorised into the following:

Respiratory diseases diagnosis. Human breath contains thousands of volatile organic compounds in gas phase. An e-nose can diagnose respiratory infections such as pneumonia. It does so by comparing smell prints from the breath of a sick patient with those of patients with standardised readings.

Urinary tract infections. An e-nose can be used as a potential diagnostic tool for patients affected with kidney diseases, by distinguishing traces of blood in urine samples.

Cancer detection. An e-nose is capable of distinguishing the breath of

a healthy person from a person with cancer. The device is promising because it can detect cancer before tumours become visible in X-rays.

Environmental monitoring.

This includes analysis of fuel mixtures, detection of oil leaks, testing ground water for odours, identification of household odours, identification of toxic wastes, air-quality monitoring and monitoring factory emissions, among others.

Application in food industry.

An e-nose has the following applications in this area:

Analysis of fruit ripening. Fruit ripening is associated with accumulation of aromatic volatile substances during ripening. An e-nose can help remove rotten fruits at the appropriate time. This can help in avoiding storage losses due to rots and fruit diseases.

JPL e-nose. This is a full-time, continuously-operating event monitor used in International Space Station. It is designed to detect air contamination from spills and leaks in the crew habitat. It can also be used to monitor cleanup processes after a leak or spill.

An e-nose can be used for detection of explosives, in space applications, research and development industries, quality-control laboratories and process and production departments.

It can also be used in the field of crime prevention and security. It has the ability to detect drug odours despite other airborne odours capable of confusing police dogs. It may also be used for bomb-detection at airports. Through careful placement of several e-noses and effective computer systems, authorities can triangulate the location of bombs to within a few metres of their location in less than a few seconds. It can also be used in the detection of harmful bacteria and detection of drug odours. **EFY**